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E-280 PANEL DESIGN

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to the of site, installation, commissioning and testing of Electrical Panels in accordance with Part 4, Subsection 1.2 Summary of Work.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards” and the following specific standards:

2.1 National

PUIL	:	Peraturan Umum Instalasi Listrik
SPLN	:	Standard Perusahaan Umum Listrik Negara.
SII	:	Standard Industry Indonesia

2.2 International Electrotechnical Commission

IEC 60028	:	International standard of resistance for copper
IEC 60044	:	Instrument transformers
IEC 60051	:	Direct acting indicating analogue electrical measuring instruments and their accessories
IEC 60059	:	IEC standard current ratings
IEC 60071	:	Insulation co-ordination
IEC 60079	:	Electrical apparatus for explosive gas atmospheres
IEC 60081	:	double-capped fluorescent lamps - Performance specifications
IEC 60083	:	Plugs and socket-outlets for domestic and similar general use standardized in member countries of IEC
IEC 60146	:	Semiconductor convertors - General requirements and line commuted converters
IEC 60228	:	Conductors of insulated cables
IEC 60245	:	Rubber insulated cables - Rated voltages up to and including 450/750 V

IEC 60255	:	Electrical relays
IEC 60269	:	Low-voltage fuses
IEC 60282	:	High-voltage fuses
IEC 60297	:	Dimensions of mechanical structures of the 482.6 mm (19 inch) series.
IEC 60309	:	Plugs, socket-outlets and couplers for industrial purposes
IEC 60331	:	Tests for electrical cables under fire conditions - Circuit integrity
IEC 60439	:	Low-voltage switchgear and controlgear assemblies
IEC 60529	:	Degrees of protection provided by enclosures (IP Code)
IEC 60598	:	Luminaires
IEC 60617	:	Graphical symbols for diagrams
IEC 60811	:	Common test methods for insulating and sheathing materials of electric cables and optical cables.
IEC 60921	:	Ballasts for tubular fluorescent lamps - Performance requirements
IEC 60947	:	Low-voltage switchgear and controlgear
IEC 61082	:	Preparation of documents used in electrotechnology

2.3 National Fire Protection Association

NFPA 70 : National Electrical Code

3. OPERATING AND DESIGN CONDITIONS

The Electrical panels shall be designed for outdoor service location or indoor for tropical climate along the seacoast. Refer to Part 4, Subsections 1.1, 2.14 and 2.16 for meteorological, seismic and environmental conditions of the site.

4. DETAILED REQUIREMENTS

4.1 General

This Technical Specification is applicable to control type panels, cabinets, and electrical enclosures.

All panels shall have the following degrees of protection as defined by IEC-60529:

- (1) Indoors in an air conditioned room: IP 42
- (2) Indoors away from fluid piping and tanks in an enclosed room: IP 42
- (3) Other indoor locations: IP 54.
- (4) Outdoors: IPW 55.
- (5) All panels located in hazardous area shall be designed in accordance with IEC 60079.

4.2 Control Power Supply

Molded case circuit breakers (MCCBs) for control power supplies shall meet the following requirements:

- (1) MCCBs shall be designed and tested in accordance with IEC.
- (2) They shall be suitably rated for both the continuous and short circuit loadings of the circuits.
- (3) For three-phase circuits, the circuit breakers shall be of the three-pole type; for single-phase circuits they shall be of the single-pole type and for DC circuits they shall be of the double-pole type.
- (4) All circuit breakers shall be provided with an auxiliary contact for remote indication of circuit breaker operation.

Fuses and links shall meet the following requirements for control circuit protection:

- (1) Fuse and link holders and bases shall be in accordance with IEC 60269 and IEC 60282, as applicable.
- (2) Fuses shall be of the non-deteriorating HRC (high rupturing capacity) type and fuse bridges shall be so designed as to permit detection of fuse-operation without removing the bridge.
- (3) Test links in current transformer circuit shall be of plug-in type.

4.3 Construction

Each panel shall be fabricated of sheet steel (ASTM), not less than 2 mm thick, to form a completely enclosed rigid, free standing structure, suitably braced,

stiffened and reinforced. The panel structure shall be fabricated of channels or angles not less than 4 mm thick. Exterior steel for housings, auxiliary equipment compartments, doors, covers and filler plates shall be cold-rolled sheet steel. Welded construction shall be used throughout.

Each panel shall not be more than 2,300 mm high, with length and depth as required.

4.4 Painting and Finishing

Painting and finishing shall be in accordance with applicable Technical Structural/Civil Specifications S-420 and S-430. The internal color shall be white (R.A.L 9010). The exterior finish color shall be light grey (R.A.L. 7030).

4.5 Doors, Openings and Covers

Access doors of non-walk-in type shall have a hinge assembly that shall permit door removal for unrestricted access to the internal equipment. All doors shall have locks, flush-mounted handles and 3-point latches.

The Contractor shall provide all necessary mounting screws and fastening devices which shall be of the rust-proof type.

Openings with gasket cover plates for cable entry shall be provided.

Insulating grommets shall be furnished and installed through openings provided in metal barriers and wiring troughs for the passage of wires or cables.

4.6 Terminal Blocks

Terminal blocks shall be heavy duty molded bakelite. Screw type terminals with washer head screws shall be provided for terminating all wiring and interconnecting cables.

Terminal blocks for use with panel wiring shall have a minimum rating of 600 V, 30 ampere. The terminal blocks shall be provided with a designation strip not less than 15 mm wide for identification of each incoming or outgoing wire.

Terminal blocks shall be provided for termination of all wiring entering or internal to the panels except for prefabricated plug-in cables. A maximum of two (2) conductors shall be terminated at any one (1) terminal point. No solder terminal connections shall be used.

For shielded field cables, a separate terminal shall be provided for each shield drain wire.

A minimum of 15% spare terminals shall be provided. Only one (1) wire per terminal will be used on the outgoing side of these blocks. Any common connections required shall be provided on the panel side of the block.

4.7 Wiring

4.7.1 Wire

All shop wiring shall be done with Class 2 or 5, as required, stranded copper wires. The connections between fixed and hinged sections shall be done with Class 6 (extra flexible) wire.

Wires shall be fire retardant polyvinyl chloride (PVC). The wire size and voltage ratings shall be in accordance with E-230.

4.7.2 Wiring Arrangement

Wires and wire groups, extending from device to device or from devices to terminal blocks, shall be installed in wireways (wire throughs) and/or packed in neatly formed bundles securely clamped or tied together with cable ties attached and secured to the panel framework with wire cleats.

Shielded instrument cables carrying low-level signals shall be in bundles or wireways separate from those for power or control cables.

4.7.3 Wiring Termination

All internal and external wiring terminations shall be made with insulated, ring type, compression (solder less) lugs.

Shielded cable drain wires and shield tapes shall be fully insulated and terminated at terminal blocks. Shields shall be grounded, at both ends.

Grounding of shield wires for ungrounded thermocouples shall be at or near the thermocouple leads terminal boxes. Instrument cable shields shall be grounded at the panel.

All incoming power terminals shall be clearly identified in a manner distinctly

different from all other termination for safety in maintenance.

The Contractor shall wire all spare contacts from each auxiliary relay and control switch, to terminal blocks.

4.8 Nameplates

Nameplates shall be installed to designate the service of all switches, instruments, meters, fuse blocks, relays, amplifiers, etc. both internally and externally located on the panels.

Any instrument which is mounted on the panel front shall have nameplates on both the front and rear of the instrument. The instrument nameplates mounted on the front of the panel shall be engraved with English language legends. A detailed nameplate list with service legends shall be submitted for the Owner's review and approval.

4.9 Panel Mounted Devices

4.9.1 General

Front-of-panel mounted devices shall not be located lower than 760 mm or higher than 1700 mm above the floor level.

Devices inside the panel shall not be located lower than 500 mm or higher than 1700 mm above the floor level.

All relays, pressure switches and other electrical devices having contacts shall have dust-tight enclosures.

All control switches, tripping and interposing relays, etc. for the circuit breakers with dual trip coils shall have sufficient contacts to provide a redundant control of each trip circuit.

4.9.2 Protection Equipment and Relays

All protection equipment and relays shall be in accordance with Technical Specification E-001.

4.9.3 Indicating Lamps

All indicating lamps shall be in accordance with Technical Specification E-001.

4.9.4 Control Switches

All control switches shall be in accordance with Technical Specification E-001.

4.9.5 Pushbuttons and Pushbuttons Stations

All pushbuttons and pushbuttons station shall be in accordance with Technical Specification E-001.

4.9.6 Instrument Transformers

All instrument transformers (current and voltage transformers) shall be in accordance with Technical Specification E-001.

4.9.7 Indicators in Panels

Dials and scales shall be white background with black letters, numerals and graduations and shall contain service identification. Letters for service identification shall be black for dials and single indicating scales and for the left hand identification of double indication scales. Letters for the right hand identification of double identification scales shall be red.

Meter ranges shall be chosen such that primary scale indication normal maximum measurement is $\frac{2}{3}$ to $\frac{3}{4}$ of full scale on the meter.

Meters shall have accuracy class 1.0. except otherwise specified.

4.10 Interior Lighting

230 V AC fluorescent interior lighting with switches shall be furnished inside the panel. The lighting shall be connected via suitable fuses directly to the incoming 230 V AC power supply. The light fixtures shall have a degree of protection against electrical shock of Class I or Class II according to IEC 60598. Class I fixtures shall be grounded.

4.11 Convenience Plug Receptacles

230 V, two-pole plug receptacles with side earthing contacts shall be furnished inside each panel. They shall conform to design of IEC 60083.

4.12 Grounding

All panels shall be equipped with a copper ground bus of not less than 7 x 50 mm cross section extending the entire length of the panel. The ground bus shall be securely anchored to the panel so as to effectively ground the entire structure and shall be connected to grounding system.

4.13 Ventilation Fans

If forced ventilation is required to dissipate heat generated internally, ventilation fans complete with intake filters and all necessary grilles shall be furnished. Each blower shall be of sufficient capacity to maintain 0.5 millibar positive pressure within the panel and limit the air temperature rise to 5°C. An alarm contact shall be provided to indicate that the intake filter is clogged.

4.14 Anti-Condensation Heaters

Panels and enclosures located in non air-conditioned areas shall be equipped with anti-condensation heaters suitable for a 230 V AC single phase 50 Hz supply. Heaters shall be automatically controlled by thermostats and shall be of corrosion-proof material, rated for 400 V AC and operated at 230 V AC.

4.15 Mimic Bus Representation

Mimic bus shall be provided on the local control panels to represent the primary equipment and their connections. The final mimic bus color coding, arrangement, representation and size shall be subject to the Owner's review.

The mimic bus shall be of polyvinyl chloride material or approved equal. It shall be glued in place and shall be secured with rust- proof machine screws.

5. PERFORMANCE TESTING AND GUARANTEES

The Contractor shall perform all type and shop tests necessary to demonstrate the suitability of his equipment. All test reports and certificates shall be subject to the approval of the Owner.

All test in according to the IEC standard of applicable equipment shall be performed. For type test, the Contractor may submit the type test certificate of identical equipment and no repeated test is required.

After erection, the Contractor shall demonstrate the proper operation of the equipment. Special attention should be given to testing the equipment in all operating modes.

6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

The Contractor shall fulfill the quality assurance/quality control requirements listed for individual equipment in the respective equipment specifications and in Part 4, Section 2 Subsection 2.7.

The quality assurance/quality control procedures shall also cover, but not be limited to, the following:

- (1) Design criteria and engineering calculations
- (2) Factory and field inspections
- (3) Wiring insulation tests
- (4) Control and functional tests
- (5) Protective relay settings and tests

The quality assurance/quality control documentation shall include, but not be limited to, the following:

- (1) Inspection Reports
- (2) Certified Test Reports
- (3) Material certification for the specified requirements of cables, insulation, conductor material, etc.
- (4) Reports of seismic testing of the panels and panel mounted equipment.

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